

Identifying Contact Fingers on Touch Sensitive Surfaces by Ring-Based Vibratory Communication

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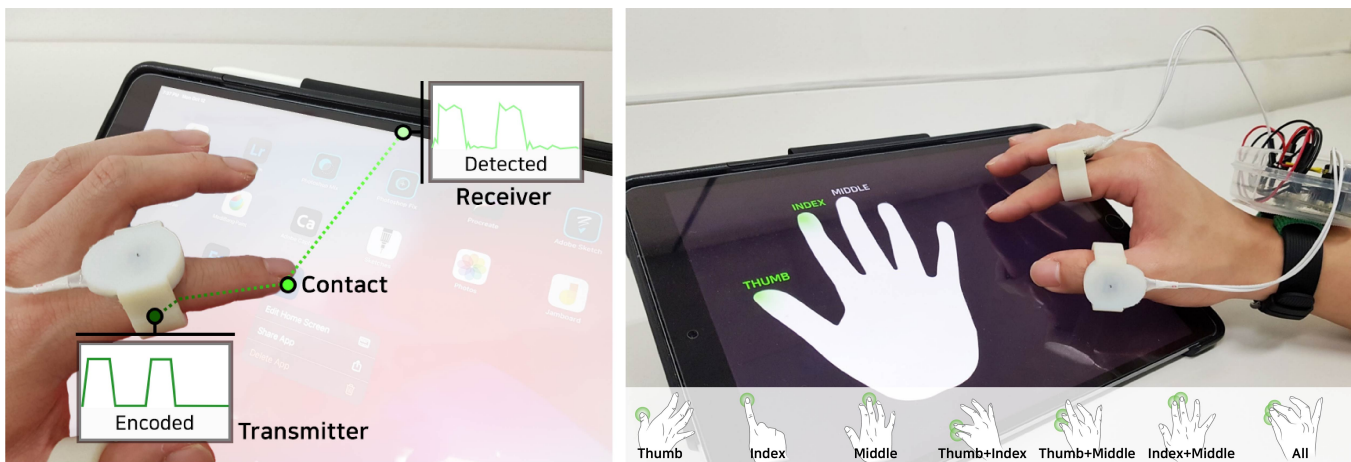


Figure 1: System overview (left). Illustration of an interaction scene and seven recognizable contact finger states (right).

ABSTRACT

As computing paradigms shift toward mobile and ubiquitous interaction, there is an increasing demand for wearable interfaces supporting multifaceted input in smart living environments. In this regard, we introduce a system that identifies contact fingers using vibration as a modality of communication. We investigate the vibration characteristics of the communication channels involved and simulate the transmission of vibration sequences. In the simulation, we test and refine modulation and demodulation methods to design vibratory communication protocols that are robust to environmental noises and can detect multiple simultaneous contact fingers. As a result, we encode an on-off keying sequence with a unique carrier frequency to each finger and demodulate the sequences by applying cross-correlation. We verify the communication protocols in two

environments, laboratory and cafe, where the resulting highest accuracy was 93 % and 90.5 %, respectively. Our system achieves over 91 % accuracy in identifying seven contact states from three fingers while wearing only two actuator rings with the aid of a touch screen. Our findings shed light on diversifying touch interactions on rigid surfaces by means of vibratory communication.

CCS CONCEPTS

• **Human-centered computing** → **Interaction devices**; *Ubiquitous and mobile computing*; • **Hardware** → **Sensors and actuators**.

KEYWORDS

Finger Identification, Vibratory Communication, Vibration Sensing, Touch Interaction

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1 INTRODUCTION

Wearable devices offer unique interaction opportunities since the devices always reside on the body [23, 36]. Devices, such as smartwatches, smart rings, and on-skin interfaces, enable control and command of other computing devices anytime and anywhere. However, the small size of the wearables inevitably compromises the efficiency and diversity of interactions [54]. This drawback limits the interaction capability of the devices and acts as a critical obstacle to seamless interaction. This study explores an input sensing method that identifies contact fingers to diversify touch interactions while wearing ring-type wearables.

Across various computing devices, there have been many previous attempts to recognize the contact finger(s) of a user. They use either implicit or explicit information. The first approach infers contact fingers from highly correlated actions or states during touch, for example, hand posture [24, 38, 46, 59], finger contact state [9, 25, 52], arrangement of the hand and device [37], and muscle tension of the arm [2, 3]. This approach generally presents relatively low classification accuracy (around 80–90 %) or limited sensing capability (2–3 single-finger contact) because of the inherent ambiguities. To overcome these limitations, researchers encode fingers with explicit markers or hardware, such as visual markers [10, 27, 28, 53], vibration actuators [4, 44], motion sensors [16, 29, 31, 49], and distance sensors [12, 13]. The explicit approach provides good accuracy, but it still has issues in recognizing multiple contact fingers or requires special-purpose hardware.

To address this technical challenge, we introduce a finger identification method capable of detecting multiple contact fingers based on data communication technology. As for the modality of data communication, we rely on a *mechanical wave*. Although mechanical vibration has a narrow bandwidth compared with other conventional communication modalities, such as radio, acoustic, and light waves, it is more easily accessible; commercial smart rings, such as Amazon Echoloon [1], often include vibration actuators to deliver haptic feedback. Moreover, vibrations propagating through the human body or smartphones can be captured by low-cost receivers, for example, a microphone [20, 40, 51] and an accelerometer [4, 23, 41]. In this study, a system for vibratory communication is constructed using easily accessible hardware and commercial devices.

As depicted in Fig. 1, the user wears rings capable of transmitting vibration. The rings serve as the transmitters and trigger encoded vibration sequences once contact is detected. As the receiver, the microphone inside the tablet records the vibrations that have propagated across the finger and device. The vibration recordings are demodulated to identify the fingers that currently touch the device's surface. Recognizing contact fingers corresponds to identifying which transmitters on the fingers have triggered the vibration sequences.

We first measured the vibration transmissibility of the communication channels involved, the hand and a touch screen, to test the feasibility of vibratory communication. Using the transmissibility data, we built a testbed to simulate the propagation of encoded vibration sequences. We extensively tested our modulation and demodulation methods to verify the effects of the design parameters by simulation. To consolidate the simulation results, we conducted a validation user study at two places, the laboratory and a cafe. The

results suggest that our system and protocols are accurate during communication and robust against sound noise, with the highest accuracies of 93 % and 90.5 %, respectively. Finally, we verified the effectiveness of our vibratory communication as a finger identification method with a touch screen. In this experiment, we included the contact states from a finger not wearing a ring. Our system can accurately identify contact fingers at both places, presenting over 91 % accuracy across seven contact states enumerated from three fingers with two rings (see Fig. 1).

The main contributions of our study include the proposal of a system and protocol for contact finger identification with special-purpose vibratory communication and the validation of its performance and robustness across different users and places. The use of vibration allows for cost-effective communication between a user and a computing device. Along with the systematic design process and data analysis, we implement our system with sensors and actuators available in commercial devices. We believe that our concept presented in this study can be beneficial for diversifying touch interaction on electronic devices with a microphone, such as IoT and wearable devices, and tangible interfaces, such as touch tracking on rigid surfaces [19, 26, 44] and touch detection on everyday objects [35, 43].

2 RELATED WORK

2.1 Contact Finger Identification

Early work on contact finger identification employs cameras along with explicit visual markers, such as color tags [10, 53] and fiducial tags [28]. Marquardt et al. [27, 28] presented a holistic toolkit for tracking and detecting the hand on tabletops and associated the toolkit with interaction techniques. As a proof-of-concept study, Goguy et al. [10] explored bimanual interaction techniques rather than technical aspects. The implication of these early studies is the design of interaction techniques, assuming successful finger identification.

Researchers have focused more on marker-free vision-based methods. Sridhar et al. [46] trained a random forest model using colored markers on the fingers. The model could recognize contact fingers without the markers from depth images, but the accuracy was lower than 80 %. Holz and Baudisch [15] captured fingerprints with a high-resolution camera to identify the contact fingers on a rear-projected touch screen. The other approach is to estimate contact fingers from the joint constraints of the hand. Zheng et al. [59] inferred the finger pressing a key from several hand skeletal features reconstructed from RGB images. Park et al. [38] utilized a fisheye camera for the same purpose, pursuing both a wide interaction space and simple hardware configuration for mobile devices, and the recognition accuracy was around 83 %. In a similar vein, capacitive images are used for contact finger recognition on a mobile device [24, 25, 52], a tabletop [7], and a smartwatch [9]. Although these methods require no additional hardware, the overall accuracy is below 90 % [9, 25], and the recognition target is limited to either single [9] or multiple [7, 52] contact fingers only. Using external hardware to sense hand grips improved the multi-finger identification accuracy to 95.8 % [24].

Other sensing methods have also been extensively studied. First, motion sensors are attached to the hand for passive sensing. Tactile

Table 1: Summary of prior work on finger identification technology.

	Hardware		Contact Finger(s)	[T,I,M,R,L]	Test Results [%]
	Surface	Body			
DeepFishEye [38]	Fisheye Cam	-	Single	[ooooo]	83.3 (Single)
Le et al. [25]	-	-	Multiple	[oo...]	84.0 (Single)
InfiniTouch [24]	Grip Sensing	-	Multiple	[ooooo]	95.8 (Multiple)
MagTouch [37]	IMU	Magnet	Single	[.ooo.]	95.0 (Single)
TapID [31]	-	Accelerometers	Single	[ooooo]	91.2 (Single)
WhichFingers [29]	-	Vibration Sensors	Multiple	[ooooo]	99.6 (Single); 95.6 (Two-Finger)
TelemetRing [49]	-	Inductive Sensing	Multiple	[ooooo]	94.4 (Single); 89.1 (Two-Finger)
Ours	Microphone	Vibration Actuators	Multiple	[ooo..]	92.5 (Single); 89.6 (Two-Finger)

*In the array ([Thumb, Index, Middle, Ring, Little]), the letters ‘o’ and ‘.’ denote recognizable and unrecognizable contact fingers.

Teacher [16] affixed three piezo vibration sensors on the back of the hand to distinguish five single contact fingers and their multi-touch combinations; the recognition accuracies were 89 % and 85 %, respectively. Attaching the same sensors to all fingers, WhichFingers [29] improved the accuracies to 99.6 % and 95.6 %, respectively. Recently, Meier et al. [31] presented the possibility of reducing the number of sensors when recognizing single contact fingers by introducing a convolutional neural network to two accelerometer data on the wrist. Takahashi et al. [49] presented a ring-shaped battery-free piezoelectric accelerometer for the hardware simplicity of ring devices. Their finger identification accuracies were 94.4 % and 89.1 %, respectively. Second, measuring the muscle tension on the forearm is another passive sensing method. Benko et al. [3] and Becker et al. [2] captured electromyography (EMG) to identify single contact fingers; however, this method suffers from low discriminability and high individual variances. Thirdly, magnetic sensing is a good option. Park et al. [37] attached a magnet to the middle finger to distinguish the touch from its two adjacent fingers and reported an accuracy of 95 % for the three single contact fingers. Last, a distance sensor attached to the fingertip led to over 99.5 % accuracy for differentiating a finger with and without the sensor [12, 13].

Mechanical vibration, if used for sensing, enables the use of frequency domain information with less computing power and simple signal processing electronics because of its relatively low-frequency range. Bianchi and Je [4] demonstrated single-finger identification on a tablet by detecting vibrations. Limited by the hardware performance, they tested only very low-frequency vibrations from 4 to 20 Hz without further perceptual or system optimization, and formal user studies. AudioTouch [22] detects pinching gestures between the thumb and the other fingers by actuating and receiving sinusoidal sweep signals (20–40 kHz) from two piezo elements on the back of the hand. VersaTouch by Shi et al. [44] was designed for finger position tracking (and pressure sensing) using vibration transmitters on the fingernail and piezoelectric receivers on a rigid surface. Additionally, they briefly tested the possibility of multiple finger identification at far separated touch positions by time-division multiplexing of 2 kHz vibration pulses. In general, increasing vibration frequency lowers tactile sensitivity but decreases the vibration transmissibility (see Section 4.2.3 and 3.3). Unlike [22, 44], our system triggers vibration only when contact

is detected. To this end, we thoroughly explore vibration-based communication for robust and reliable protocols using mid-range vibrations (0.5–0.7 kHz) and then apply those results to finger identification. We support the simultaneous recognition of multiple contact fingers using different carrier frequencies for each finger wearing an actuator in a natural position.

We summarize the previous studies on finger identification in Table 1. Our method requires users to wear rings on each finger. In general, identifying multiple contact fingers requires external hardware (i.e., grip sensing [24]) or mechanisms to explicitly encode each finger (e.g., vibration sensors [29], coils [49], and vibration actuators [44]). Among the previous approaches, attaching a motion sensor to each finger has presented the best performance, and a product using this approach has been commercialized [50]. Compared to the approach, our method requires additional hardware to be attached to a target surface, but it offers unique opportunities to quickly prototype and test contact finger identification with commercial products.

2.2 Vibratory Data Communication

Mechanical (or physical) vibrations have received a growing interest as a means of data communication. Compared to other conventional means of communication, such as radio, acoustic, and light waves, mechanical vibrations are more affordable [56, 60] and secure [17, 21, 60].

Researchers have presented vibratory data communication using rigid propagation media, such as wood and metal. Hwang et al. [17] demonstrated the feasibility of secure data transmission between two smartphones using their built-in vibration motor and accelerometer. Ripple [41] improved the transmission rate of data communication by introducing advanced modulation and demodulation techniques. VibroComm [56] achieved an even higher transmission rate by employing gyroscopic sensors and allowing direct physical contact between a receiver smartphone and a vibrating surface.

There is also a rich literature on vibratory communication that considers the human body as a propagation medium. OsteoConduct [60] first showed that body-area vibration communication is possible through bone conduction. ViBand [23] investigated several modulation schemes for body-mediated vibratory communication

by using an accelerometer of a smartwatch as the receiver. An interesting application reported by Kim et al. [21] used vibration to connect to medical devices implanted in the body. A setup similar to our system, a ring-type vibration transmitter on the finger and a microphone receiver, was considered in Ripple II [40] for touch-based authentication, but Ripple II allowed direct contact with the receiver.

Most of the previous approaches tested data transmission across a fixed [17, 41] or variable [21, 23, 40, 60] single transmission medium. In our case, the medium used for vibration transmission was much more complex. First, the medium involves multiple fingers, each of which presents different vibration propagation dynamics, which are also user-dependent. Second, the medium includes a touchscreen device. Each device has unique vibration propagation dynamics depending on its mechanical design, which is also dependent on the transmission path. Hence, we deal with multiple transmission media with substantially different characteristics, which also vary with users and contact positions. Instead of transmitting complex data sequences, we examine the feasibility of vibratory communication as an interaction technology. Our system identifies contact fingers across a surface by simultaneously triggering multiple ring-type transmitters.

3 CHARACTERIZATION OF COMMUNICATION CHANNELS

Our vibration transmission consists of two sequential phases: the hand to the fingertip and the fingertip to a device. For successful data transmission, a vibration needs to travel through a series of channels. Therefore, we first investigated the vibration characteristics of the channels involved, that is, the hand and the surface of a tablet. From these experiments, we characterized the parameters and physical constraints that affect the quality of vibration transmission and determined the range of potential carrier frequencies for system implementation.

3.1 Apparatus

As a vibration transmitter, we use a surface transducer (PUI Audio ASX02108R) connected to a 3.1 W mono audio amplifier (Texas Instrument TPA-6211A1EVM). To measure the vibrations transmitted through the hand, we use a contact microphone (Knowles BU-27173). An internal microphone records vibro-acoustic signals in a tablet (ASUS Nexus 7). The transmitter and receiver communicate with a sampling rate of 44.1 kHz through a data acquisition card (National Instruments USB-6251) for the contact microphone case and a sound card (Realtek ALC887) for the tablet.

3.2 Estimating Vibration Transmissibility

To quantify the ability of each channel to transmit mechanical vibration, we identify its frequency response over a wide frequency range using the exponential sine sweep method [8]. The actuator produces a 1.5-s long input signal $x(t)$, a 1-s vibration with the frequency varying exponentially from 0.3 to 3 kHz, and a subsequent 0.5-s silence. The receiver synchronously records the vibration that has passed through the channel.

Using the collected data, we first estimate the impulse response $h(t)$. According to a previous study [8], we can estimate the impulse

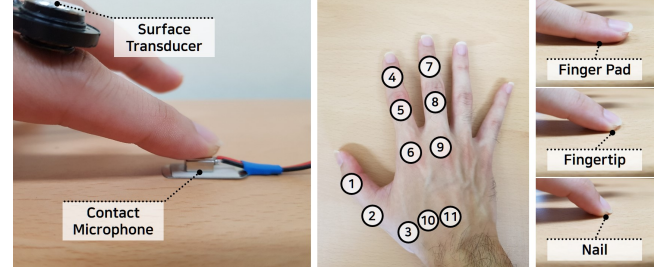


Figure 2: Our experimental settings for the vibration transmissibility of the hand (left). 11 transducer positions over the hand (middle) and three contact poses (right).

response of the channel by the convolution of the recorded output signal, $y(t)$, and the time reversal of the transmitted input signal, $f(t) = x(-t)$, such that $h(t) = y(t) \otimes f(t)$. We then compute the channel frequency response, $H(f) = \mathcal{F}\{h(t)\}$, by applying fast Fourier transform (FFT) to the estimated impulse response. $|H(f)|$ represents the transmissibility of vibration for the tested frequency range.

3.3 Transmissibility of Vibration: Hand

We collected vibration transmission data from the hand to the fingertip (Fig. 2, left) for different actuation positions and contact poses. The vibration transducer was affixed to one of the 11 positions (Fig. 2, middle): the beginning of the phalanges (middle and proximal for the *index* and *middle* finger; proximal for the *thumb*) in each finger, and the beginning and end of the metacarpals in each finger. Data were captured under three contact poses: *finger pad*, *fingertip*, and *nail* (Fig. 2, right). The contact microphone was used for vibration sensing to ensure that the signals were propagated through the body rather than the air [20, 51]. In total, we collected 33 signals (11 positions $\times$ 3 contact poses) from five participants.

The average frequency responses were computed across the participants, and their magnitude responses are shown in Fig. 3. As expected, the vibration transmissibility was degraded as the frequency increased. At all measurement positions, *finger pad* resulted in the lowest transmissibility. From our pilot tests, a vibration that presented less than -160 dB gain in Fig. 3 was not reliably detected by our system. When an actuator is close to the fingertip (position 1, 4, 7, 5, and 8), *fingertip* and *nail* can transmit sufficient vibration energy (> -140 dB) at frequencies ranging from 0.5 to 1 kHz (shaded areas in Fig. 3). Thus, we mainly consider *fingertip* in the following sections to reflect ordinary touch interactions. The transmissibility of *thumb* was generally lower than that of the other two fingers, presumably because of its larger size and mass.

The observations above suggest that position 1, 5, and 8 are the optimal positions for mounting the surface transducers. The transmitted signal intensities were balanced across the three fingers. It has also been reported that the three positions are subjectively preferred when placing ring-type sensors on the finger [11].

3.4 Transmissibility of Vibration: Tablet

For the second phase of data transmission, we selected a tablet (ASUS Nexus 7) as a testbed touchscreen with a microphone. The

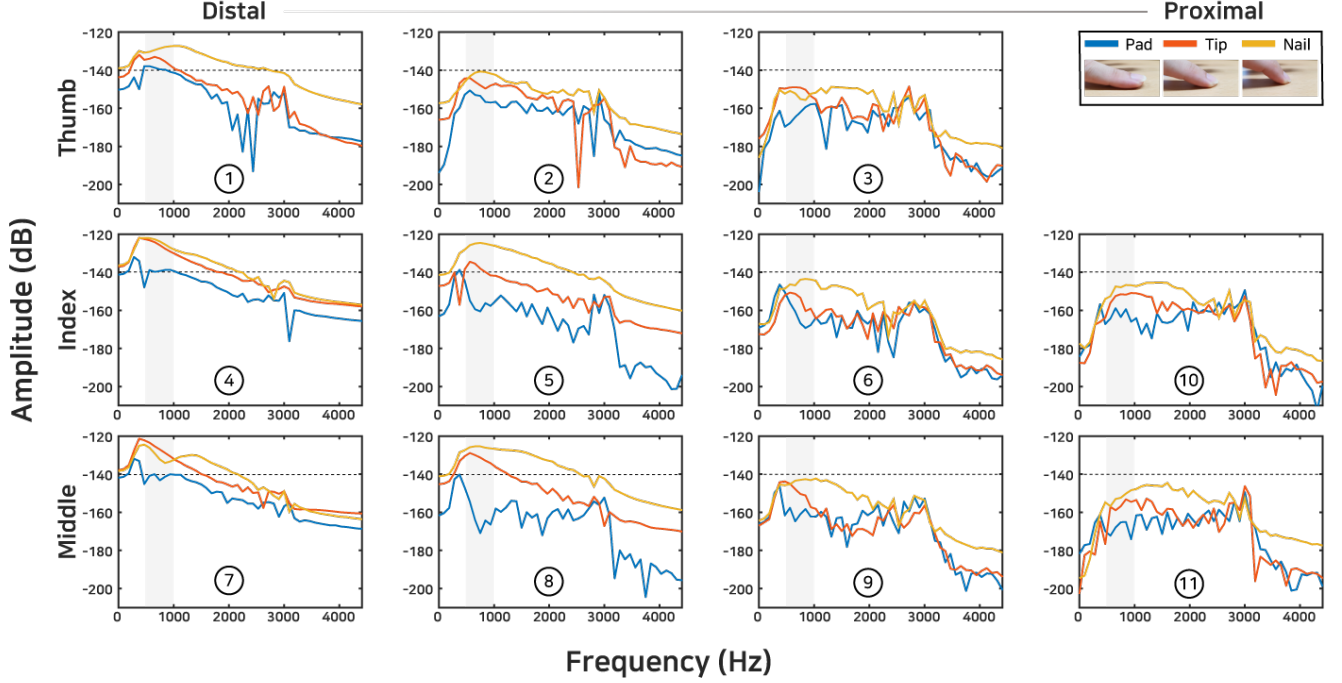


Figure 3: Magnitude responses of the hand. The black dotted lines denote -140 dB gain in each plot, and the shaded areas represent the frequency range from 0.5–1 kHz.

microphone pinhole was covered with double-sided tape to reduce external noise and prevent the detection of airborne propagation [20]. We collected signals at six equally spaced grid positions on the tablet (Fig. 4, middle).

The measured magnitude responses are presented in Fig. 4. The vibration transmission gains were generally degraded when the vibration-applied position is farther from the microphone located on the top left side. Other regular patterns were not observed, which may be due to the complicated mechanical interactions between the parts in the tablet. The vibration transmissibility is acceptable for frequencies lower than 700 Hz, regardless of the vibration-applied positions.

4 SYSTEM

4.1 Principle of Operation

Our system includes two ring-shaped vibration transmitters and a touch surface with a microphone receiver. The user wears two rings in two fingers. An encoded vibration emitted by one finger travels through the finger and then the surface. Only the vibration(s) from the contact finger(s) reaches the receiver, and the contact finger(s) can be identified.

On-off keying (OOK), a well-known modulation technique for its simplicity, conveys digital data by turning on and off a carrier signal and makes a sequence of pulses for transmission [17, 21, 58]. In this study, we modulate and transmit OOK sequences with different carriers to notify the contact fingers. The modulation scheme is

designed and tested to achieve robust performance in noisy environments by repeating and lengthening a unit OOK sequence.

To demodulate the encoded sequence, we use cross-correlation, which measures the similarity between two sequences. If an input signal to the transmitter is cross-correlated with a recording at the receiver, the point where the cross-correlation is maximized is likely to indicate the existence of the encoded sequence within the recording. Further details are provided in the rest of this section.

4.2 Modulation

4.2.1 Hardware. We designed wearable ring-type hardware for transmitting vibrations, as shown in Fig. 5. This hardware serves as a transmitter for vibratory communication. To make the hardware both comfortable to use and adaptable to individual differences, we selected elastic and bendable material (Stratasys Tango Plus [47]) and made two pairs of different sizes (Fig. 5). The hardware affixed the actuator on the finger, and its vibrating plate maintained direct contact with the skin. We used the same surface transducer and audio amplifier as described in Section 3.1. The actuator has a resonance frequency of 640 ± 128 Hz, which guarantees reliable vibration generation in the frequency range to be tested (500–724 Hz).

For the receiver, we utilized the internal microphone of the tablet with its hole covered and used this receiver setup in our main experiments. This low-cost receiver effectively captured vibro-acoustic signals without direct attachment to the tablet surface. Second, we demonstrated the microphone of a wireless earbud (QCY T1s) by attaching it directly to everyday surfaces with double-sided

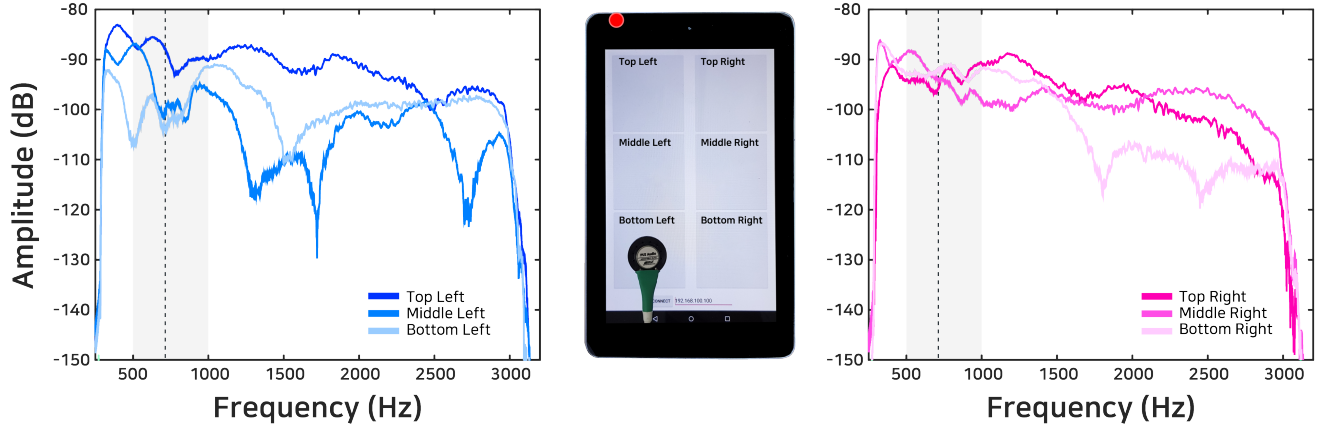


Figure 4: Magnitude responses of a Nexus 7 tablet. A red dot denotes its microphone position. Black-dotted lines indicate 0.7 kHz, and shaded areas represent the frequency range from 0.5–1 kHz. The six data acquisition positions are illustrated within the device picture.

tape. This reconfigurable setup allows fast prototyping of the finger identification on rigid plain surfaces.

4.2.2 Signal. The OOK sequence tested in our system is formulated by the following three parameters: pulse count p_c , pulse duration p_d , and carrier frequency f_c , such that

$$s_{pulse}(t) = \begin{cases} a_{envelope}(t) \sin(2\pi f_c t) & 0 \leq t < p_d/2 \\ 0 & p_d/2 \leq t < p_d \end{cases}, \quad (1)$$

$s_{pulse}(t)$ is repeated p_c times in a single OOK sequence. To prevent the audible noise from spectral leakage [14], we apply a Hamming window, $a_{envelope}(t)$, to the beginning and end of the pulse in the 10 % of p_d [51].

An OOK sequence generates unique patterns when it is cross-correlated with other OOK sequences that have different carrier frequencies. The patterns are determined by the pulse duration and the carrier frequency difference. These patterns are illustrated further in Section 4.3, and more plots are available in the supplementary material.

In our pilot tests, we first encoded different carriers using a single pulse; however, this method did not effectively differentiate the pulse from environmental noise. The current communication scheme is much more robust against noise because we can observe the two unique patterns (Fig. 7(c) and (d)) only when two encoded OOK sequences have a specific carrier frequency difference and share the same pulse duration.

4.2.3 Carrier Frequency Selection. The channels of our system include the human body. Selecting an appropriate carrier frequency is highly important because it determines the quality of the signal transmission and also the perceptibility of the carrier vibration. We assigned different frequency pairs to each of the three two-finger combinations, *thumb-index*, *thumb-middle*, and *index-middle*, for the best design. Here, we focused on the three most frequently involved fingers during touch interaction with wearable devices [9, 33].

We considered the perceptual sensitivity of both auditory and cutaneous sensory channels. The human vibrotactile sensitivity is

the highest at around 250 Hz [32, Figure 2], and it degrades as the frequency increases [32]. Humans cease to perceive vibrations if the frequency is approximately 1 kHz [48]. In contrast, the auditory sensitivity improves as the frequency increases until 1 kHz [18, 32]. To minimize tactile sensations, we selected carrier frequencies higher than 500 Hz. To reduce auditory sensations, we increased transmission and transmissibility balance between a finger pair.

We determined the specific carrier frequencies for each two-finger combination based on their vibration transmission. Fig. 6 illustrates the transmissibility curves of the three fingers on the tablet by combining their frequency responses. Because the index finger showed the lowest transmissibility, we first selected 612 Hz as its carrier frequency, around its local maximum after 500 Hz. For the thumb and index finger pair, we assigned 500 Hz to the thumb, where 112-Hz frequency difference was found to be the best trade-off between vibration transmissibility, perceptual sensitivity, and demodulation accuracy in pilot experiments. Similarly, for the middle and index finger pair, we selected 724 Hz as the carrier frequency for the middle finger. For the thumb and middle finger pair, we first assigned 596 Hz to the thumb from the thumb’s local maximum and then 708 Hz to the middle finger.

4.3 Demodulation

In this section, we describe the detection and decoding of the vibration sequences from the transmitter. Our signal demodulation process consists of three steps: preprocessing, detection, and decoding.

We denote the input sequence to the vibration transmitter i by $s_i(t)$ and the signal recording at the receiver by $r(t)$. In preprocessing, the two input and recorded output signals are cross-correlated. The results are normalized and then fed to the Hilbert transform to obtain the envelope of the normalized cross-correlation, $e_i(t)$. If $r(t)$ (Fig. 7(a)) includes the carrier frequency of $s_i(t)$, $e_i(t)$ shows



Figure 5: Our Hardware for the transmitter and receiver. A user wears two ring-type transmitters (left). A closer look at the flexible rings (middle). A microphone of two commercial devices, a tablet and an earbud, is tested as the receiver (right).

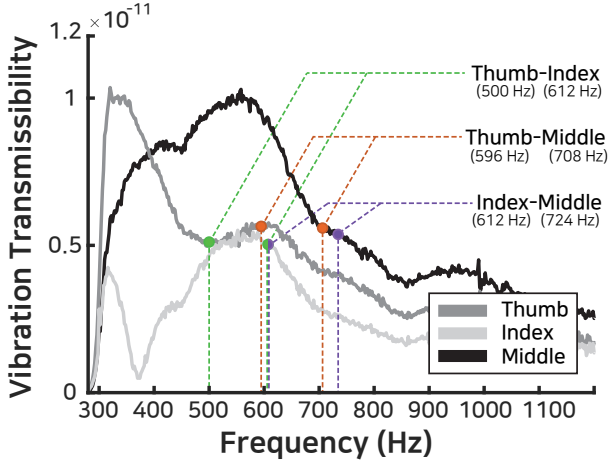


Figure 6: The combined vibration transmissibility of the three fingers on the tablet. The colored circles and dotted lines indicate the carrier frequencies assigned to two fingers for each pair, thumb-index (green), thumb-middle (orange), and index-middle (purple).

prominent main and side peaks that are separated by p_d , as illustrated in Fig. 7(c). If $r(t)$ includes another carrier frequency, $e_i(t)$ includes many narrow peaks, as shown in Fig. 7(d).

The next detection step aims to judge whether any signal has been transmitted. Thus, we compute the average of the two input signals $s_1(t)$ and $s_2(t)$, denoted by $s_{avg}(t)$, of different carrier frequencies. We then find the cross-correlation between $r(t)$ and $s_{avg}(t)$ and its envelope function $e_{avg}(t)$. Then we retrieve the highest peak in $e_{avg}(t)$ such that

$$e_{avg}^{max} = \max e_{avg}(t) \text{ and } t^{max} = \operatorname{argmax} e_{avg}(t). \quad (2)$$

We also obtain the average value of the two side peaks by

$$e_{avg}^{side} = \frac{1}{2} (e_{avg}(t^{max} - p_d) + e_{avg}(t^{max} + p_d)). \quad (3)$$

If $r(t)$ includes any carrier frequency components of $s_1(t)$ or $s_2(t)$, e_{avg}^{max} is $2 \cdot e_{avg}^{side}$ in the ideal case. Such a unique pattern is very unlikely to occur unless the OOK sequence of the same carrier

frequency is used as the input, and this observation can be a basis for robust decision rules.

Then, we define the ratio between the main and side peaks by

$$L = \frac{e_{avg}^{max}}{e_{avg}^{side}}, \quad (4)$$

and consider a decision statistic D , assuming a normalized Gaussian distribution for L , such that

$$D = \frac{L - \bar{L}}{\sigma(L)}, \quad (5)$$

where $\bar{L}$ is the mean of L , and $\sigma(L)$ is the standard deviation. We use $\bar{L} = 2.008$ and $\sigma(L) = 0.095$, which are estimated through extensive simulations that also include the effects of external noise (Section 5). For a 95.0 % correct decision, we judge that an input signal is detected in $r(t)$ if $|D| < 1.96$ and move to the next step.

In the last decoding step, we inspect from which transmitter on the finger the encoded sequence had emanated. Here we only consider the subsequence $\tilde{r}(t)$ of $r(t)$, such that

$$\tilde{r}(t) = \begin{cases} r(t) & \text{if } t \in [t^{max}, t^{max} + p_d p_c], \\ \tilde{r}(t) = 0 & \text{otherwise.} \end{cases} \quad (6)$$

For each i , we perform (2)–(5) using $s_i(t)$ and $\tilde{r}(t)$ instead of $s_{avg}(t)$ and $r(t)$. We then declare that $\tilde{r}(t)$ includes the carrier frequency of $s_i(t)$ and so does $r(t)$ if $|D| < 1.5$ and $e_i(t^{max})$ is greater than all its neighbors in $(t^{max} - p_d/2, t^{max} + p_d/2)$. The former criterion encompasses the 86.6 % confidence interval to reject more false positives, while the latter checks whether $e_i(t^{max})$ is indeed the greatest peak to rule out the misleading cases shown in Fig. 7(d). This decoding step is simple and effective, enabled by our encoding rule that generates very distinct patterns. This efficiency in the demodulation process improves the practical applicability of our system when applied to a variety of contexts without training classification models.

5 SIMULATION

We performed extensive simulations to optimize the modulation and demodulation scheme. The simulation pipeline is shown in Fig. 8. For a finger i wearing a vibration transmitter, the input signal $s_i(t)$ is transformed by FFT to $S_i(f)$. This input signal spectrum $S_i(f)$ is multiplied by the frequency response $H_{finger,i}(f)$ of the

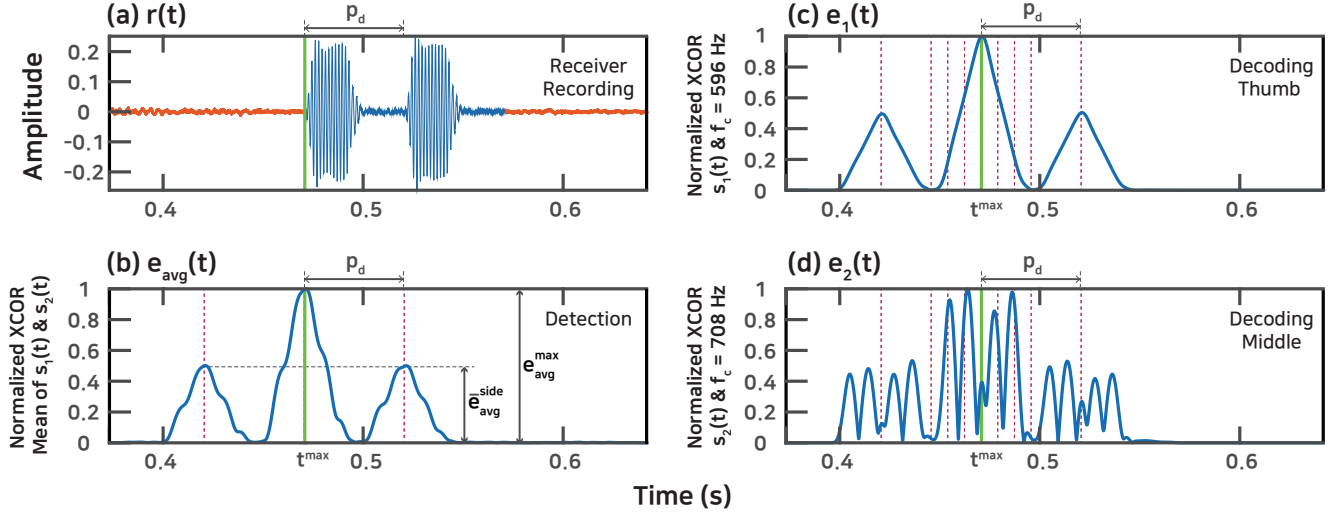


Figure 7: The visualization of data processing. (a) A sample recording when a user touches a tablet with the thumb ($p_c = 2$, $p_d = 0.05$ s, $f_c^{thumb} = 596$ Hz and $f_c^{middle} = 708$ Hz). (b) The cross-correlation envelope $e_{avg}(t)$ for detecting the encoded sequence in the recording. The green lines denote t^{max} , and the height of the main peak (e_{avg}^{max}) is $p_c/(p_c - 1)$ times the average height of the side peaks (e_{avg}^{side}). (c) and (d) respectively indicate $e_i(t)$ for the decoding step. The red dotted lines indicate whether $e_i(t^{max})$ is the maximum within $(t^{max} - p_d/2, t^{max} + p_d/2)$.

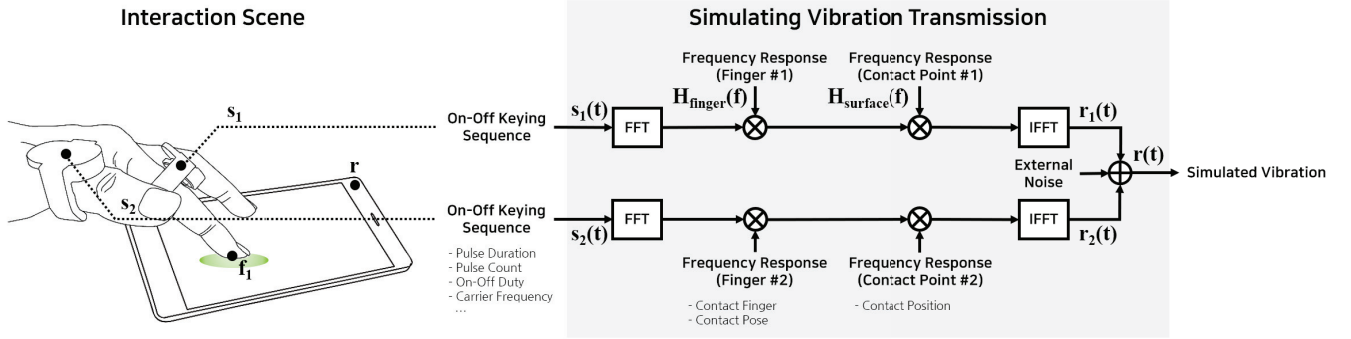


Figure 8: Block diagram for simulating the transmission of vibration across a surface.

finger and then by the frequency response $H_{surface,j}(f)$ at the contact position j on the tablet. $H_{finger,i}(f)$ and $H_{surface,j}(f)$ were obtained from the data collected in Section 3. The result is transformed by the inverse FFT to a time-domain response $r_i(t)$. For the simulation of two-finger contact, $r_1(t)$ and $r_2(t)$ are computed for the two fingers, and they are added to $r(t)$: the simulated output to be measured by a microphone. For robust parameter optimization, external noise is injected into $r(t)$.

The specific goal of the simulation was to find the best values for pulse count p_c and pulse duration p_d . As a constraint, the duration of one sequence should be less than 200 ms, reflecting that the perceptual thresholds that users notice a response delay in GUI range from 150 to 200 ms [5]. Performance measures were: *Accuracy*—the

correct rate of our demodulation process and *Onset Error*—the absolute time difference between the time index at which an encoded sequence is detected and the ground truth.

One batch of the simulation consisted of 14580 trials under the following conditions: 1) pulse duration $p_d = 25, 50, 75$ ms; 2) pulse count $p_c = 2, 4, 6, 8$ for $p_d = 25$ ms, $p_c = 2, 3, 4$ for $p_d = 50$ ms, and $p_c = 2, 3$ for $p_d = 75$ ms; 3) four noise conditions, imitating the sound noises collected from four different locations in our university: *laboratory* (average = 37.3 dB; peak = 59.6 dB), *outside* (48.4 dB; 72.0 dB), *hallway* (49.0 dB; 78.7 dB), and *cafe* (54.5 dB; 68.4 dB), all Z-weighted; 4) five scale factors for each noise signal; and 5) 81 contact conditions, mixing no contact, 12 single-finger contact cases (2 fingers $\times$ 6 touch positions), and 36 multi-finger cases (6 touch positions for one finger $\times$ 6 for the other). These three cases were

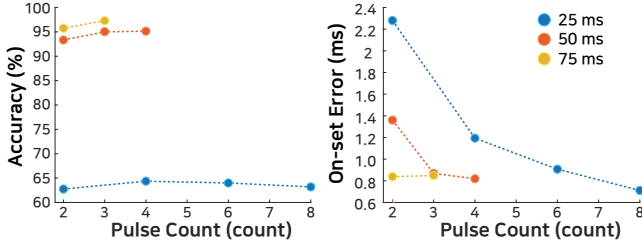


Figure 9: Simulation results: Accuracy (left) & Onset Error (right).

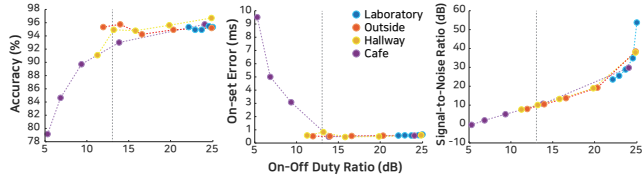


Figure 10: Simulation results of $(p_d, p_c) = (50 \text{ ms}, 2)$. The black dotted lines indicate the critical OOR level (13 dB).

repeated by 9, 3, and 1 times, respectively, for balanced parameter optimization.

Each batch of the conditions was repeated three times for each of the three ring configurations, *thumb & index*, *thumb & middle*, and *index & middle*. As a result, the simulation proceeded with more than 130 k trials. To represent the noise level in each trial, we also computed two ratios of the energy between signal and noise: signal-to-noise ratio (SNR) and on-off duty ratio (OOR), both in dB.

Fig. 9 presents the average values of *Accuracy* and *On-set error* for different combinations of p_d and p_c . At $p_d = 25 \text{ ms}$, *Accuracy* was unacceptably low (below 65%), and $p_d = 25$ was not further considered. At $p_d = 50$ and 75 ms , *Accuracy* was high over 93%, and increasing p_c (more pulse repetitions) tended to improve *Accuracy*. *Onset Error* was also good with values lower than 1.4 ms.

We examined the simulation results more carefully for the following three signal parameter pairs: $(p_d, p_c) = (50 \text{ ms}, 2)$, $(50 \text{ ms}, 3)$, and $(75 \text{ ms}, 2)$, where all the signal durations were short and no more than 150 ms. The results were very similar regardless of the signal parameter pairs. In Fig. 10, we show the results for $(p_d, p_c) = (50 \text{ ms}, 2)$ to assess the effect of noise. The remainders can be found in the supplemental material.

Fig. 10 demonstrates that our noise conditions were effective in expressing a wide range of noises, covering SNR from 0 to 55 dB (the y-axis range of the right panel) and OOR from 5 to 25 dB (the x-axis range). Note that SNR increases monotonically with OOR (Fig. 10, right). To obtain high *Accuracy* over 92% (Fig. 10, left) and low *Onset Error* below 0.8 ms (Fig. 10, middle), OOR should be greater than 13 dB (marked by vertical dotted lines), where SNR is 10 dB. This result suggests that our communication scheme remains effective until SNR is decreased to 10 dB, as evidence of its robustness to external auditory noise.

6 EXPERIMENT

The next step was conducting an experiment in which our vibratory communication scheme was tested by users in two places: *laboratory* (the best noise condition) and *cafe* (the worst), for the three signal conditions: $(p_c, p_d) = (50 \text{ ms}, 2)$, $(50 \text{ ms}, 3)$, and $(75 \text{ ms}, 2)$, now denoted by *50 ms*×2, *50 ms*×3, and *75 ms*×2, respectively. We also tested the three ring configurations: *thumb & index*, *thumb & middle*, and *index & middle*. We were mainly concerned with contact recognition accuracy under different conditions. All experiments reported in this paper were conducted under protocols approved by the Institutional Review Board at the authors' institution (PIRB-2019-E018).

6.1 Methods

We recruited 10 right-handed participants (10 male; 23.9–29.6 years). The participants wore ring-type transmitters on two fingers, as shown in Fig. 5. They touched the tablet in Fig. 4 with the designated finger(s) at random positions. This contact was detected by the tablet, and the encoded vibration sequences were sent to the transmitters. A 1-s-long signal was captured concurrently by the tablet microphone. The participants were instructed to maintain stable light contact in a relaxed hand posture while avoiding finger pad contact until the vibration generation was completed.

The experiment consisted of three sessions: adaptive gain setting, practice, and test sessions. In the adaptive gain setting session, we adjusted the power of the transmitters for each participant and testing place. Participants touched the tablet center ten times, using the index finger while wearing the rings on their index and middle fingers. During this process, the experimenter increased the transmission power from zero to the maximum and then measured the OOR values. These values were fitted to a linear function to determine the power sufficient at that moment in that place. Then, we set the transmission power to ensure average OOR across the participants to be 18.6 dB at *laboratory* and 15.3 dB at *cafe*. The corresponding signal amplitudes were 35–55% and 51–84% of the maximum, respectively.

In the practice session, participants touched the tablet in all contact combinations of the index and middle fingers for each signal condition. The main test session consisted of 9 sets (3 signal conditions × 3 ring configurations). The presentation order of the signal conditions was randomized first, after which the order of the ring configurations was randomized for each signal condition. Each set included 20 trials (5 single touches each and 10 multi-touches) in random order. Between the sets, participants had a one-minute break and then re-wore the rings for the next set. All participants completed the 180 trials in each of the two places, and it took approximately 90 min in total. Finally, we asked participants to rate the perceived intensity and unpleasantness of sound and vibration on a 7-point Likert scale.

6.2 Results and Discussion

Fig. 11 illustrates the mean accuracy of our system for all signal conditions, places, and ring configurations. The highest accuracy was 93% at *laboratory* (*50 ms*×3 and *thumb & middle*) and 90.5% at *cafe* (*50 ms*×2 and *thumb & middle*). At both places, *thumb & middle* presented the highest accuracy, followed by *thumb & index* and *index*

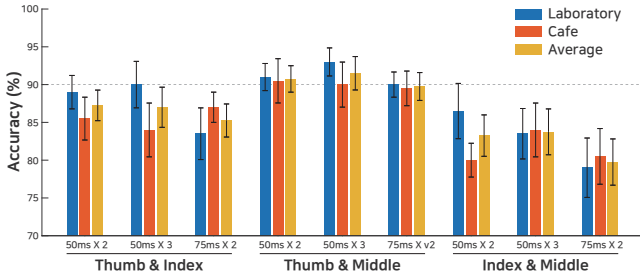


Figure 11: Mean accuracy of all communication protocols and systems tested at the two places. The error bars represent standard errors.

& middle. To clarify the effects of the tested variables, we performed a three-way repeated measures ANOVA. The only significant main effect was the ring configuration ($F(2, 18) = 9.35; p = .002$). The place ($F(1, 9) = 1.16; p = .309$) and the signal condition ($F(2, 18) = 2.41; p = .118$) were insignificant. According to a Tukey's HSD test, *thumb & middle* presented significantly higher accuracy than the other two. No difference was found between *thumb & index* and *index & middle*. We observed a significant interaction only between the signal condition and place ($F(2, 18) = 110.56; p = .018$). At *laboratory*, a Tukey's HSD test showed that the accuracy of $50\text{ ms} \times 2$ and $50\text{ ms} \times 3$ was statistically higher than that of $75\text{ ms} \times 2$. However, at *cafe*, we found no significant differences between the signal conditions.

According to the above statistical tests, our system achieved a similar decoding performance, regardless of the tested places. This was despite the considerably larger and variable noise levels at the *cafe*. This result may also indicate that the gain setting was appropriate. The mean and standard deviation of OOR across the participants was 22.3 ± 1.7 dB at *laboratory* and 14.1 ± 2.3 dB at *cafe*, which correspond to the expected ranges in Section 6.1.

Our system achieved the best decoding performance when the ring-type transducers were affixed to the thumb and middle fingers. One reason for this is that it is impossible to fully block the propagation of vibration toward the other fingers. The *thumb – middle* configuration offers the most distant transducer positions and can mitigate vibration propagation most effectively.

Regarding the signal conditions, $50\text{ ms} \times 3$ presented the highest average accuracy for most ring configurations. Thus, $50\text{ ms} \times 3$ is recommended when the primary concern is the accuracy of a system. If a system has constraints in power consumption or transmission time, it is reasonable to choose the shorter encoding, $50\text{ ms} \times 2$, as $50\text{ ms} \times 2$ showed comparable performance to $50\text{ ms} \times 3$. When comparing $50\text{ ms} \times 3$ and $75\text{ ms} \times 2$, repeating the shorter pulse more times was beneficial.

Participants' responses to the perceived intensity and unpleasantness of sound and vibration were reported later in Section 8.1.

7 SUMMATIVE EVALUATION

The main objective of this study is to implement an interaction technology for finger identification by introducing communication schemes for vibratory data communication. Thus far, we have designed and verified the communication protocol and system. In

this section, we report on a user study conducted to test the performance of our system for finger identification. We also tested the possibility of detecting contact states that include a finger not wearing a ring.

7.1 Methods

We recruited 10 participants (7 males and 3 females; 21.3–27.8 years) for this user study. One participant was ambidextrous and used the right hand during the experiment. None of the participants participated in the previous experiment.

We used the best-performing conditions ($50\text{ ms} \times 3$ and *thumb & middle*) and average signal amplitudes (40 % at *laboratory* and 60 % at *cafe*) reported in the previous section. The participants put ring-type transmitters on their thumb and middle fingers. We included the index finger without rings because of its frequent usage [33]. Therefore, there were seven contact states: *thumb*, *index*, *middle*, *thumb+index*, *thumb+middle*, *index+middle*, and *all*. We asked the participants to touch the touch screen of the tablet showing the six button areas as in Fig. 4. The other instructions and settings were the same as those in the previous experiment.

Here, we inferred the contact states from two information sources, detected carrier frequencies and the number of touch points. For example, if the thumb carrier frequency and two touch points are detected, the system receives it as *thumb & index*. The decision rules are shown in Fig. 12. We observed that our system infrequently detected two weak carrier frequencies from the index finger in the pilot tests, and such cases were assigned to *index*. We additionally checked the symmetricity of the adjacent side peaks in the demodulation to reduce false positives.

All participants completed two sessions, first at the *cafe* and then at the *laboratory*. There were 42 trials (6 touch areas $\times$ 7 contact states) for a single set. We included a practice set at the beginning, and the participants repeated two sets in each session. The order of the contact states was randomized first, and the order of the touch areas was randomized for each contact state. The participants had a one-minute break after each set, and then they re-wore the rings. The experiment took up to 75 min. After finishing each session, we asked them to rate the subjective questionnaires in Section 6.1.

7.2 Results and Discussion

Fig. 12 shows the confusion matrix for each place. Across the participants, the average accuracy was 91.1 % (SD=3.6 %) at *laboratory* and 91.4 % (SD=2.7 %) at *cafe*. We applied two-way repeated measures ANOVA with the independent factors of place and contact state. We found no difference between the two places ($F(1, 9) = 0.07; p = .794$). There was a significant effect of the contact state on the recognition accuracy ($F(6, 54) = 7.38; p < .001$), but the interaction between the two factors was not significant ($F(6, 54) = 0.85; p = .538$). These results confirm that our system can recognize contact fingers at both noisy and quiet places. Therefore, we continued our data analysis by pooling the data of the two places.

The recall and precision for all the contact states are shown in Fig. 13. On average, recall was 91.3 % (SD=6.0 %), and precision was 93.7 % (SD=6.0 %). The most noticeable contact state was *index+middle*, where the recall was only 79.6 %. According to a

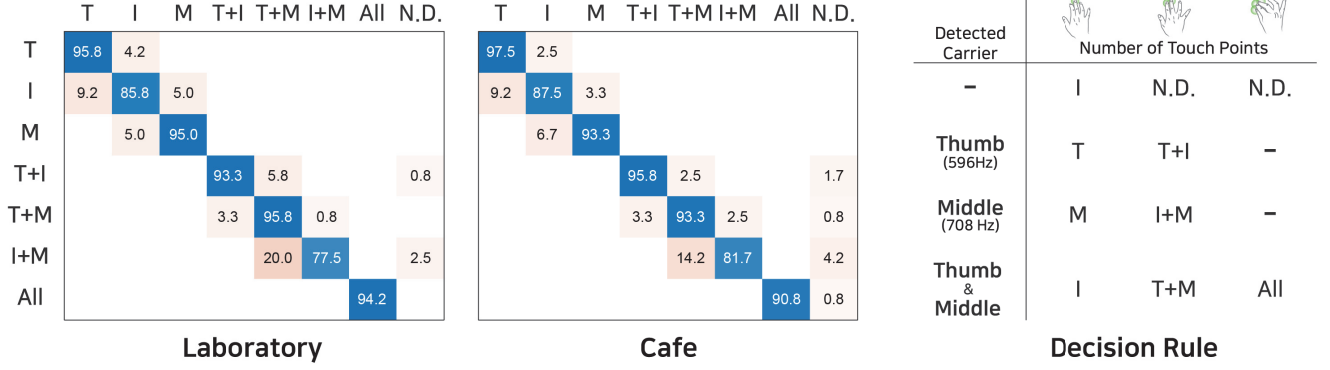


Figure 12: Confusion matrices in percentage for the two tested places, *laboratory* and *cafe* (left). Our decision rule for contact finger identification (right). Note that triple finger touches detected with a single carrier frequency were regarded as wrong instances.

Tukey’s HSD test, the accuracy of *index+middle* contact was statistically lower than that of *thumb*, *middle*, *thumb+middle*, and *all*, and no such differences were found between the other contact states. We observed that 24 out of 49 incorrect instances were from the middle left side of the tablet. Second, the recall for *index* was 86.7 %. We found 17 out of 32 errors from the middle left and bottom left sides of the tablet. Excluding *index+middle* improved the statistics to 93.2 % (SD=3.5 %) and 95.2 % (SD=3.3 %). These results suggest that our system can effectively identify at least six contact states.

As illustrated in Fig. 4, the transmissibility of the middle finger carrier frequency (708 Hz) is relatively low on the left side of the tablet because of its mechanical design and internal structure. This was not problematic for *middle* and *thumb+middle*, which included only the fingers wearing the ring. However, additional weak signals from the finger without the ring increased recognition errors. We found no such effect from the thumb carrier frequency (596 Hz).

In this user study, we investigated the feasibility of our communication scheme in different places and users; however, a few other factors can affect the performance of our system. The first is the distance from the receiver to the contact point. As the distance increases, the quality of the received signal generally decreases. Therefore, it is essential to specify the critical level of OOR required for good performance. We observed that OOR and accuracy were highly correlated ($r = 0.90; p < .001$). Our system achieved good accuracy ($> 93\%$) after a 13 dB OOR, which corresponds to approximately 7 cm from the receiver.

Second, the contact force and pose can affect the vibration transmission. A greater contact force is generally beneficial for transmitting vibrations (> 200 Hz) within the finger [42] owing to the tight coupling between the finger and the contact surface. In our test, increasing the contact force improved the amplitude of encoded vibrations at the receiver, and even an extreme contact force (5 N) did not significantly degrade the signal amplitude. Finally, the fingertip and nail poses resulted in a decent signal quality even at a weak contact force (around 0.5 N), and the pad pose presented relatively low signal quality. We believe that our system has no significant problems during casual touch interactions because low finger pitch postures are relatively difficult to perform and uncommon [30].

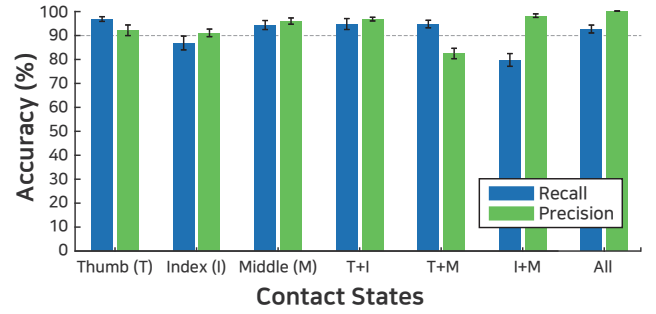


Figure 13: Recall and precision for seven contact states. The error bars represent standard errors.

Participants’ responses to the perceived intensity and unpleasantness of sound and vibration were reported later in Section 8.1.

8 GENERAL DISCUSSION

From a series of experiments, we have demonstrated the feasibility of identifying contact fingers on a touch screen by introducing vibratory communication. We hope that our attempt will accelerate the application of vibratory communication in wearable devices.

8.1 Vibration and Human Perception

A common issue in sensing technologies that utilize mechanical vibration is that users may feel vibration and sound [4, 6, 34, 35, 44, 57]. Therefore, we carefully designed the vibration sequences to minimize the sensations. The vibrotactile sensitivity of the carrier frequencies we used (> 500 Hz) is approximately 15–70 times lower than that of the most sensitive frequency (around 200 Hz) [32]. We also attempted to determine the optimal signal intensity and duration through experiments. We minimized the intensity by introducing the minimum OOR required for successful demodulation and reduced the signal duration to 100 ms.

In Fig. 14, we report the participants’ subjective ratings of vibration and sound collected in the two user experiments. On average,

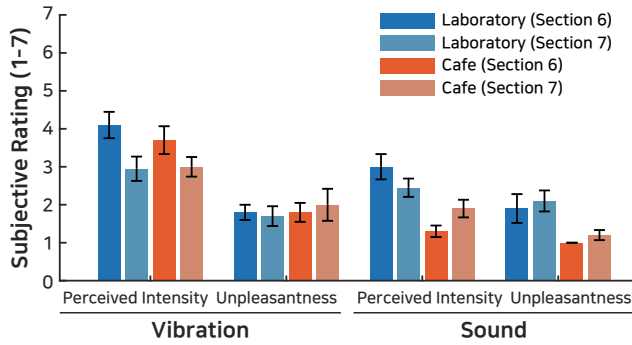


Figure 14: Mean the subjective ratings collected in the two user experiments. The error bars denote standard errors.

they rated vibrations as moderate (score 4) to slightly weak (3) and sounds as moderately weak (2). The unpleasantness was rated from very low (1) to moderately low (2) for both vibration and sound. Interestingly, both the unpleasantness and perceived intensity of sound decreased at the cafe, although the actual signal intensities were higher than in the laboratory. As future work, we are interested in elaborating vibration sequences so that users can accept them as natural sensory feedback.

8.2 Real-World Applicability

There are several real-world constraints for delivering vibrations from the transmitter to the receiver. The first is environmental noise. In the first experiment, our system presented good decoding performance in both quiet and noisy places by calibrating the signal intensity per place, and we used the data to optimize the signal intensity of the subsequent experiment. For the best performance, it will be beneficial to devise automatic adaptive gain tuning by employing advanced data communication techniques such as preamble.

As for vibratory communication, the human body is not an ideal channel, exposed to factors introducing variability such as individual differences, poses, and movements [40]. In the summative experiment, we partially verified that our system and protocol performed well with a fixed signal intensity across the participants using the calibration data of the first experiment. However, we did not experimentally test or control the effect of contact force and pose, and we conducted the experiments under relatively stationary contact conditions. We also note that the best carrier frequency may depend on the person. To minimize this effect, we selected carrier frequencies from the average impulse response of the hand across participants.

The applicability of our system and protocol to other devices and surfaces has not been widely explored or explicitly tested. Therefore, we measured the distribution of OOR using the identical wireless receiver (Fig. 5) attached to four different devices. Our system detected encoded vibrations with acceptable signal quality, and the average OORs were 11.6 (17-inch Monitor), 18.4 (iPhone 12 mini), 18.9 (5.7-inch Trackpad), and 19.1 dB (Nexus 7) across their surfaces. Detailed results are provided in the supplementary material. We agree that a reliable working area might not be very large on heavy

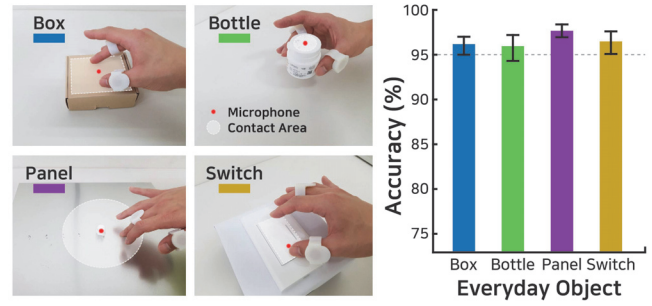


Figure 15: Illustration of four everyday objects and respective receiver positions (left). Mean finger identification accuracies of our system (50 ms×3, thumb & middle, and 40% intensity in the laboratory) for the four objects (right).

surfaces (e.g., a desk and the monitor), but even a small area around the receiver is beneficial because the area can function as a finger-aware button. We also conducted a supplemental study of everyday objects by attaching the wireless receiver (Fig. 15). For ten participants, we asked them to touch four everyday objects in different contact states (single-touch, multi-touch, and two-finger pinching). The mean identification accuracy of 1400 trials was 96.4%. Scalability to larger everyday objects and surfaces will be one of our future research topics.

8.3 Hardware Improvement

Our current implementation is not mobile and requires a wired connection via an audio port. As such, its design can be improved using wireless audio IC chips. Currently, two rings require lower power consumption (< 10 mW) to trigger the encoded signals than the maximum support of commercial wireless audio systems (e.g., Apple Airpod supports max 100 mW). In addition, the computation for demodulation is fast enough (< 8 ms), but we expect a delay of at least 150 ms in total (50 ms × 2 signal; 40 ms AptX-II latency [39]).

In this study, we focus on contact finger identification, not on detecting touch or sensing other aspects. Previous work has introduced various interaction technologies on rigid surfaces, for example, detecting initial contact [43, 45], localizing taps [19, 26, 55], and tracking continuous touch position [44], using sound and vibration. Our contact finger identification can make touch interaction on everyday surfaces more efficient and diverse in conjunction with the listed tracking technologies. It will be an interesting future direction to design an advanced data communication system that enables the tracking and recognition of fingers on rigid surfaces.

9 SCENARIOS AND APPLICATIONS

First, we illustrate the potential scenarios with our system. The benefits of our communication scheme include scalability and flexibility. A user can wear rings on their preferred fingers, or even more than two rings. Our demodulation process requires only simple parameter changes. The use of multiple rings is not limited to one hand or a single user. Two rings can be worn on different users for multi-user interaction or different hands to support bimanual interaction. We also implemented our system using the receiver and

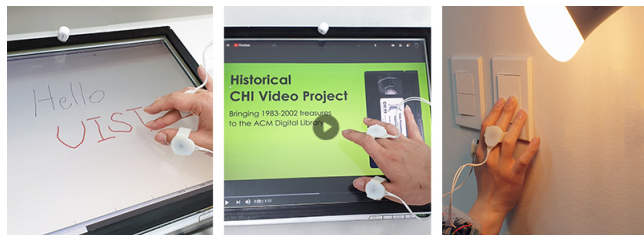


Figure 16: Illustration of our applications. Changing stroke width in our drawing application (left), finger-aware video player (middle), and finger-dependent light switch (right).

transmitter available in commercial devices. The high affordability of our system makes it an appropriate candidate for application to diverse everyday contexts with ease.

We have created applications for finger-aware touch interaction with our system. Here, we attached a wireless earbud directly to everyday objects, a monitor and a light switch, to capture vibrations from its internal microphone. Identifying contact fingers can be used to differentiate touches, such as triggering different hidden commands in a menu bar or tool bar [4, 10] and using one finger as a modifier key [59]. We implemented these in a drawing application (Fig. 16, left). The user can select the desired brush hidden in the brush button. After drawing, it is possible to change properties such as color, width, and position by touching the strokes with different fingers and then sliding. Contact finger identification can be used to execute preassigned commands [2, 29], for example, window snapping and video player controls (Fig. 16, middle). Our approach can also be used to diversify the inputs of IoT and tangible interfaces, such as a light switch (Fig. 16, right). In our application, lighting conditions such as intensity, color, and dimming rate can be mapped to different fingers. For example, turning on a switch with the thumb and middle fingers results in brighter lighting than turning on with the thumb.

10 CONCLUSIONS

In this work, we have proposed a system that enables robust and accurate finger identification on touch-sensitive surfaces by utilizing vibratory communication. We systematically design and implement our communication method and algorithm to enable the system with only off-the-shelf components and commodity devices. Our system has been validated across different users and places. We also have demonstrated the possibility of contact finger identification on small objects and surfaces by attaching a microphone directly to rigid surfaces or objects. We envision that the practical applicability of our system can enrich touch interactions in various computing contexts and devices.

ACKNOWLEDGMENTS

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